

Practical No. 1

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Roll No. :- 02

```
In [1]: import pandas as pd
        from sklearn.model_selection import train_test_split
```

```
In [5]: df = pd.read_csv("student_dataset.csv")
        print(df.head())
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	\
0	23.5	84.2	Low	Low	
1	19.7	63.6	Medium	High	
2	24.4	98.4	Medium	Medium	
3	29.6	88.6	Medium	High	
4	19.1	91.5	High	Medium	

	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	\
0	Yes	7.4	50.3	Medium	
1	Yes	6.7	83.1	Low	
2	Yes	8.0	63.5	Medium	
3	Yes	6.3	57.5	Medium	
4	Yes	6.2	66.0	Medium	

	Internet_Access	Tutoring_Sessions	Final_Exam_Score
0	Yes	1	54.3
1	Yes	2	59.4
2	Yes	0	63.8
3	Yes	2	61.0
4	Yes	2	60.7

```
In [7]: print(df.info())
        print(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6607 entries, 0 to 6606
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Hours_Studied                        6607 non-null   float64
1   Attendance                          6607 non-null   float64
2   Parental_Involvement                6607 non-null   object
3   Access_to_Resources                 6607 non-null   object
4   Extracurricular_Activities          6607 non-null   object
5   Sleep_Hours                        6607 non-null   float64
6   Previous_Scores                     6607 non-null   float64
7   Motivation_Level                    6607 non-null   object
8   Internet_Access                     6607 non-null   object
9   Tutoring_Sessions                   6607 non-null   int64
10  Final_Exam_Score                    6607 non-null   float64
dtypes: float64(5), int64(1), object(5)
memory usage: 567.9+ KB
None
```

	Hours_Studied	Attendance	Sleep_Hours	Previous_Scores	\
count	6607.000000	6607.000000	6607.000000	6607.000000	
mean	20.472726	79.979068	7.185273	74.716059	
std	5.988729	11.547717	1.290241	14.451962	
min	1.100000	60.000000	4.000000	50.000000	
25%	16.500000	70.000000	6.300000	62.100000	
50%	20.500000	79.800000	7.200000	74.600000	
75%	24.500000	90.200000	8.100000	87.400000	
max	44.000000	100.000000	10.000000	100.000000	

	Tutoring_Sessions	Final_Exam_Score
count	6607.000000	6607.000000
mean	1.488119	60.400333
std	1.216175	8.739400
min	0.000000	29.700000
25%	1.000000	54.300000
50%	1.000000	60.400000
75%	2.000000	66.600000
max	8.000000	95.200000

```
In [21]: df['result'] = df['Final_Exam_Score'].apply(lambda x: 1 if x >= 40 else 0)
```

```
In [23]: y = df['result']
```

```
In [27]: X = df.drop('result', axis=1)    # replace with your target  
y = df['result']
```

```
In [31]: X = df[['Hours_Studied', 'Attendance']]    # example  
y = df['result']
```

```
In [33]: X_train_80, X_test_20, y_train_80, y_test_20 = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
In [35]: X_train_70, X_test_30, y_train_70, y_test_30 = train_test_split(X, y, test_size=0.3, random_state=42)
```

```
In [37]: X_train_60, X_test_40, y_train_60, y_test_40 = train_test_split(X, y, test_size=0.4, random_state=42)
```

```
In [39]: print("80-20 Split:", X_train_80.shape, X_test_20.shape)  
print("70-30 Split:", X_train_70.shape, X_test_30.shape)  
print("60-40 Split:", X_train_60.shape, X_test_40.shape)
```

```
80-20 Split: (5285, 2) (1322, 2)
```

```
70-30 Split: (4624, 2) (1983, 2)
```

```
60-40 Split: (3964, 2) (2643, 2)
```

```
In [41]: import torch
```

```
X_train_tensor = torch.tensor(X_train_80.values, dtype=torch.float32)  
y_train_tensor = torch.tensor(y_train_80.values)
```

```
In [43]: X_train_keras = X_train_80.values  
y_train_keras = y_train_80.values
```

```
In [ ]:
```

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